

Cell Cycle Training App

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Context

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Basic Agent
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PhysiCell Constants

List of cells

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Default microenvironment

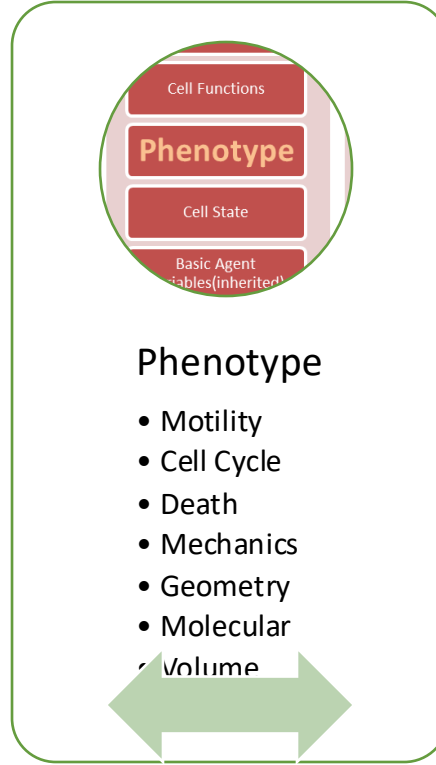


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Context



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A note about time steps

- PhysiCell is designed to account for the multiple time scales inherent to these problems, and has 3 time scales:

▪ $\Delta t_{\text{diffusion}}$	diffusion, secretion, and uptake	(default: 0.01 min)
▪ $\Delta t_{\text{mechanics}}$	cell movement	(default: 0.1 min)
▪ Δt_{cell}	phenotype and volume changes	(default: 6 min)
- This allows some efficiency improvements: not all functions need to be evaluated at each time step.
- See the PhysiCell method paper. (Oddly, not in the User Guide (yet).)



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Cell phenotype

- One of the most critical data elements in a PhysiCell Cell is ***phenotype***
- Hierarchically organize key behavioral elements:

- ✦ Phenotype

- ♦ **cycle**: advancement through a cell cycle model
- ♦ **death**: one or more types of cell death
- ♦ **volume**: cell's volume regulation
- ♦ **geometry**: cell's radius and surface area
- ♦ **mechanics**: adhesion and resistance to deformation ("repulsion")
- ♦ **motility**: active motion (other than "passive" mechanics)
- ♦ **secretion**: both release and uptake of chemical substrates. Interfaces with BioFVM
- ♦ **intracellular**: used for interacting with intracellular models. See Sessions 10-11.
- ♦ **molecular**: a place to store internalized substrates

Documentation: User Guide, Sec. 10



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Phenotype: Cycle

- Each agent's **phenotype** had a **cycle** with:

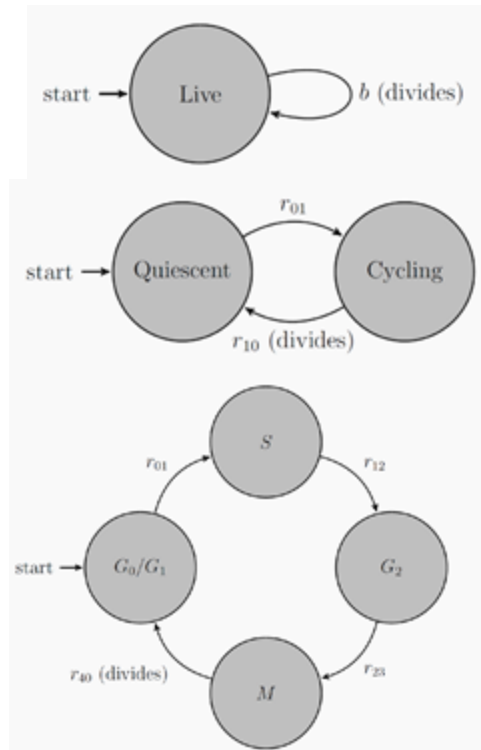
- **Cycle model**

- ♦ A directional graph: *nodes* are cycle **phases** $\{P_i\}$ and *edges* are **transition rates** $\{r_{ij}\}$
- ♦ r_{ij} is the transition rate from phase P_i to phase P_j
- ♦ One of the transitions must be marked as a *division transition*
- ♦ Users can attach **arrest condition** functions to these transitions (e.g., size checks)

- **Cycle data**

- ♦ stores the cell's current transition rates

- **Documentation:** User Guide, Sec. 11.1



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Phenotype: Cycle

Cell Cycles available in PhysiCell

- ✦Live
- ✦Ki-67 Basic
- ✦Ki-67 Advanced
- ✦Flow Cytometry
- ✦Flow Cytometry Separated
- ✦Cycling-Quiescent



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Cell definition: cycle

```
<cycle code="5" name="live">  <!-- pick a code to match PhysiCell_constants.h -->
  <!-- use phase_transition_rates OR phase_durations -->
  <phase_transition_rates units="1/min">
    <rate start_index="0" end_index="0" fixed_duration="false">0.002</rate>
  </phase_transition_rates>

  <!-- use phase_transition_rates OR phase_durations -->

<!--
  <phase_durations units="min">
    <duration index="0" fixed_duration="false">500.0</duration>
  </phase_durations>
-->
</cycle>
```

- For some problems, it's easier to work in terms of transition rates. Use the "phase_transition_rates" code for these.
 - In this example, the "live" cell cycle (with a single phase) transitions at a rate of 0.002 1/min.
- Sometimes, it's easier to work in terms of how long a cell spends in a phase. Use "phase_durations" for these.
 - In this example, the "live" cell cycle (with a single phase) lasts (on average) 500 minutes.



Background

- Cells experience a series of phases and transitions within their lifetime
- These phases comprise the **cell cycle**
- In PhysiCell, a cell cycle has 2 primary components: **phases** and **phase links**
 - **Phase**: a defined state that a cell exists in during the cell cycle
 - **Phase Link**: a transition between two phases
- Users can define their own cell cycles by creating their own phases and links
- PhysiCell also contains 6 built-in cycle models



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Phase Variables

- **Index:** the phase's unique identifier within the cycle model
- **Code:** the phase's unique global identifier within the entire PhysiCell coding environment
 - Use the constants defined in the file `PhysiCell.constants.h`
- **String Name:** the name of the phase written as a string
- **Division at Phase Exit:** indicates whether a cell should divide at the end of the phase
- **Removal at Phase Exit:** indicates whether a cell should be removed at the end of the phase
- **Entry Function:** pointer to a function executed at the beginning of the phase



Phase Links

- **Start Phase Index:** the index of the starting phase before the transition
- **End Phase Index:** the index of the phase following the phase transition
- **Fixed Duration:** indicates whether a certain amount of time passes before transitioning between phases
- **Arrest Function:** points to a function that only allows the phase transition to occur if a user-defined condition is true. Returns true if the transition is blocked (arrested)
- **Exit Function:** pointer to a function executed at the end of the phase transition



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Credits

Lesson Planning: Paul Macklin

Slides: Aneequa Sundus, Kali Konstantinopoulos*, Mary Chen*, Furkan Kurtoglu

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- Breast Cancer Research Foundation
- Jayne Koskinas Ted Giovanis Foundation for Health and Policy
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JAYNE KOSKINAS
TED GIOVANIS

Foundation for
Health and Policy



NATIONAL
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BCRF



Training materials:

* Administrative supplement to NCI U01CA232137 (Year 2)



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